# KULBIR SINGH AHLUWALIA

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## EDUCATION

**Ph.D. in Computer Science**, University of Illinois at Urbana-Champaign, USA (June 2022–Present), GPA: 3.91/4
Advisor: Prof. Katherine Driggs-Campbell (since Nov 2026).
**M.Eng. in Robotics**, University of Maryland, College Park, USA (Aug 2019–May 2021), GPA: 3.88/4
**B.Tech. in Electrical Engineering**, Punjab Engineering College, India (Aug 2015–May 2019), GPA: 8.12/10

## RESEARCH PAPERS

- **Ahluwalia, K.S.***; Jain, C.*; Cuaran, J.; Gummadi, S.; McGuire, M.; Sivakumar, A.; Hockenmaier, J.; Chowdhary, G. WaypointGen: Learning Grounded Waypoint based Navigation for Mobile Manipulators. Coordinated Science Laboratory Student Conference (CSL SC) 2026, Machine Learning, Generative AI, and Signal Processing Session. (*equal contribution)
- Cuaran, J.; **Ahluwalia, K.S.**; Koe, K.; Uppalapati, N.K.; Chowdhary, G. Active Semantic Mapping with Mobile Manipulator in Horticultural Environments. 2025 IEEE International Conference on Robotics and Automation (ICRA), pp. 12716–12722, doi:10.1109/ICRA55743.2025.11127547.
- Rangwala, M.; Liu, J.; **Ahluwalia, K.S.**; Ghajar, S.; Dhami, H.S.; Tracy, B.F.; Tokekar, P.; Williams, R.K. DeepPaSTL: Spatio-Temporal Deep Learning Methods for Predicting Long-Term Pasture Terrains Using Synthetic Datasets. Agronomy 2021, 11, 2245 (Special Issue: AI and Agricultural Robots).
- Liu, J.; Rangwala, M.; **Ahluwalia, K.S.**; Ghajar, S.; Dhami, H.S.; Tokekar, P.; Tracy, B.F.; Williams, R.K. "Intermittent Deployment for Large-Scale Multi-Robot Forage Perception: Data Synthesis, Prediction, and Planning". IEEE Transactions on Automation Science and Engineering, vol. 21, no. 1, pp. 27–47, 2024.

## PAPERS UNDER REVIEW AND IN PREPARATION

- **Ahluwalia, K.S.**, et al. WaypointGen: Compositional Trajectory Generation by Grounding Hierarchical Language Abstractions and Constraint Phrases for Outdoor Navigation. Submitted to the CoRL 2026 Workshop on Compositional Generalization for Real-World Robot Learning (non-archival); under review.
- **Ahluwalia, K.S.**; Jain, C.; Kahlon, G.S.; Cuaran, J.; Lazebnik, S.; Chowdhary, G.; Hockenmaier, J. TaP: Trajectory as Policies for Spatial Reasoning using Hierarchical Abstractions & Grounded Constraint Phrases for Mobile Manipulator Navigation. IEEE Transactions on Robotics (T-RO), manuscript in preparation.

## WORK EXPERIENCE

**Product Management Intern, Qualcomm Technologies, Inc. (Compute Chipset), San Diego, CA**
*June 8–Aug 21, 2026*

- On-site PM role on Snapdragon X Elite / X2 Elite AI-PC silicon.
- Market/competitive analysis for on-device AI/VLM/generative AI features (positioning Snapdragon X vs. Apple M-series, Intel Lunar Lake, AMD for Copilot+ AI-PC).
- Synthesized feedback into product requirements/roadmap; technical eval of VLM workloads on silicon (power/thermal/latency on Hexagon NPU).
- Cross-functional work with engineering (AIMET, AI Hub Models); internal briefings on edge AI trends.

**AI Intern, Earthsense Inc., Urbana, IL, USA** *May 2025–Aug 2025*

- Developed a natural language conditioned waypoint generation pipeline for 2D and 3D motion planners for enhancing outdoor robot navigation.
- Created an automatic labeling pipeline for large outdoor robot navigation datasets using Grounded SAM2, streamlining data processing.
- Deployed and integrated open-source Visual Language Models (Molmo-7B-D, Gemma-3-27B, Qwen-2.5-VL-72B, Qwen3-30B-A3B, Llama4-Scout, Spatial-VLM; 3B–72B params) for robot reasoning in image space and open-world NL-conditioned QA for 4-wheeled skid-steer outdoor robots.

**Teaching Assistant, CS 440/ECE 448: Artificial Intelligence, UIUC** *Fall 2026*

- Course taught by Prof. Margaret Fleck.

**Teaching Assistant, CS498GC: Mobile Robotics for Computer Scientists, UIUC** *Fall 2025*
Instructor: Dr. Girish Chowdhary.

- Co-developed the graduate course curriculum (ROS 2, sensor fusion, SLAM, mobile manipulation); authored and maintained the public course website and assignments; held weekly office hours and mentored students on robotics software.
- Recorded and published TA reference runs for the SLAM-ing Mars assignment (Husky + UR3 in the Mars Gazebo world, SLAM Toolbox occupancy map), Oct 2026.

**Teaching Assistant, CS519: Scientific Visualization, UIUC** *Summer 2025*

- Communicated complex data-visualization concepts to a multidisciplinary student audience and guided project work.

**Teaching Assistant, CS444: Deep Learning for Computer Vision, UIUC** *Jan–May 2024, 2025, 2026*
Instructor: Dr. Svetlana Lazebnik.

- Updated assignment starter code and assessed student submissions using Canvas.
- Provided support during in-person office hours and via Campuswire, addressing student queries and concerns.
- Designed and developed multimodal (Reasoning LLM/VLM proof) quiz questions, including single-choice, multiple-choice, and matching formats for students.

## RESEARCH EXPERIENCE

**University of Illinois, Human-Centered Autonomy Lab** *Nov 2026–present*
Ph.D. researcher. Advisor: Prof. Katherine Driggs-Campbell.

**University of Illinois, Distributed Autonomous Systems Lab** *Aug 2022–Dec 2023*
Graduate Research Assistant. Mentors: Dr. Girish Chowdhary & Dr. Julia Hockenmaier.

- Investigated 3D semantic representations for robot environment mapping and relocalization after failure.
- Categorized recovery action sequences for various failure cases and constructed topological maps for visual language navigation.

**University of Illinois, Distributed Autonomous Systems Lab** *Aug 2022–Dec 2023*
Graduate Research Assistant. Mentors: Dr. Girish Chowdhary & Dr. Julia Hockenmaier.

- Fine-tuned CodeT5 for natural language grounding and code generation.
- Created datasets for grounding natural language commands and analyzed location data distributions.
- Developed a Python package for the farmbot agricultural robot.

**University of Maryland, Robotics Algorithms & Autonomous Systems Lab** *Jul 2020–Aug 2021*
Independent Study. Mentor: Dr. Pratap Tokekar.

- Processed point clouds from LiDAR-mounted quadcopters for pasture simulation.
- Automated gazebo world construction for grass pastures with unique plant poses.

**University of Waterloo, Ontario, Canada** *Mar–Jul 2018*
Visiting Scholar. Mentor: Dr. Simarjeet Saini.

- Developed an orange sweetness detector using scaled conjugate gradient backpropagation.
- Designed, programmed and 3D printed low-cost photonic devices, including a urea-in-milk detector and a fundus eye camera.

**Indian Institute of Technology, Roorkee** *Jun–Jul 2016*
Research Intern. Mentor: Dr. Dharmendra Singh.

- Investigated the effect of radio wave absorbers on Radar Cross Section using Ansys HFSS.

## RESEARCH ARTICLES AND ABSTRACTS

- Cuaran, Jose; **Ahluwalia, Kulbir Singh**; Koe, Kendall; Chowdhary, Girish. "Active Semantic Mapping with Mobile Manipulator in Horticultural Environments", 2024, accepted at the 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40).
- The multispectral Fundus Eye camera prototype was featured in *Optics and Photonics News* (OPN), February 2019: Saini, Simarjeet Singh; Sridhar, Aneesh; and **Ahluwalia, Kulbir**. "Smartphone optical sensors." *Optics and Photonics News* 30, no. 2 (2019): 34–41.

## CONFERENCE PRESENTATION

- Yuqi Li, **Kulbir S. Ahluwalia**, and Simarjeet S. Saini. "Reinforcement learning integrated with supervised learning for training of near infrared spectrum data for non-destructive testing of fruits." In Sensing for Agriculture and Food Quality and Safety XII, vol. 11421, p. 114210J. International Society for Optics and Photonics, 2020.

## TECHNICAL SKILLS

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| **Languages** | Python, MATLAB, C++ |
| **Analysis & Design** | Simulink, Blender, AutoCAD, Solidworks, Ultimaker Cura, Adobe Premiere Pro |
| **Libraries** | Torch, Transformers, Numpy, Pandas, OpenCV, SciPy |
| **Other** | ROS, Gazebo, RViz, CUDA, Linux, Docker, Git, Doxygen, Anaconda, LaTeX |

## PROJECTS

- **Enhancing Stereo Depth Maps through RGBD-Conditioned Generative Models**: conditional diffusion model fusing RGB and noisy depth for metric maps; fine-tuned Stable Diffusion V2 (similar to the Marigold training pipeline)

using SimSense simulation to generate depth data; outperforms Marigold/Depth-Anything-V2 on IRS, VKITTI, NYUv2 by up to 46% Abs Rel.
- SLAM from 2D LiDAR data using split-and-merge line extraction algorithm.
- State estimation using Extended Kalman Filter for GPS+IMU+Encoder sensor fusion.
- Processed data from RTK-GPS, IMU and encoders to plot the trajectory of a field robot.
- Autonomous Vaccine Delivery Robot.
- Image segmentation using superpixels.
- Persistent-Monitoring using Multi-Robot (UAV–UGV) Coordination.
- Optimized a GestureGAN for resource-constrained settings.
- Self-adjusting roadmaps - navigation in unknown environments using LD-PRM.
- Estimated the motion of a car using Visual Odometry.

## COURSEWORK

**Graduate (UIUC):** CS598SHW: 3D Vision, CS598YL: Deep Learning for Robotic Manipulation, Advanced NLP (CS546), Machine Learning (CS446), Deep Learning for Computer Vision (CS444), Natural Language Processing (CS447), Mobile Robotics (CS498GC), Autonomous Systems and Robots.

**Graduate (UMD):** Autonomous robotics, Decision making for robotics, Visual Learning and recognition, Planning for Autonomous Robots, Perception for Autonomous Robots, Control of Robotic systems, Robot modelling, Robot programming, Building Robot software systems.

**Undergraduate (PEC):** Neural networks and fuzzy systems, Advanced control systems, Microprocessors and interfacing, Power electronics, Mechatronics, Engineering analysis and design, Manufacturing, Biomedical engineering, Electromagnetic theory, Python Programming.

## LEADERSHIP AND TEACHING EXPERIENCE

- **Excellent communication skills** demonstrated as a UIUC Teaching Assistant (CS440, CS498GC, CS444, CS519) and through graduate curriculum co-development, technical workshops, and public science outreach (featured in *Optics & Photonics News*).
- Served as **Technology Head** for the Hardware domain of the IEEE PEC Student Branch. Conducted workshops on making a "Pick n Place Transporter Robot" and "Using the Raspberry Pi" to share our team's experiences and techniques with juniors in PEC.
- Taught Math and Science to government high-school students as part of the "PUNARKRITI Welfare Society" (Jan–Apr 2016) and "Junior Einstein" (Dec 2018) social welfare organizations.

## OTHER ACHIEVEMENTS AND AWARDS

- Awarded the **National Bal Shree Award in Creative Scientific Innovations** by the Ministry of Human Resource Development, Govt. of India, conferred by the President of India - national recognition for **high levels of creativity and problem-solving**.
- **First Prize** in the MAJOR PROJECT of the B.Tech. Examination in Electrical Engineering (2015–19) for "Teleoperated Gesture-controlled Robotic Arm". [May 2019]
- Received a Certificate of Appreciation for contributions to IEEE PEC. [Aug 2017, Aug 2018]

## ADDITIONAL PROJECTS

- AR-Tag detection - superimposed an image and virtual cube on an AR tag.
- Tracked moving objects using a Lucas-Kanade Tracker.
- Baxter transporting cubes in Gazebo.
- Implemented A* algorithm for Path Planning on Turtlebot 3.
- Path planning for point and rigid robot using Dijkstra's Algorithm.
- Lane detection and Turn prediction for a self-driving car.
- Agile Robotics for Industrial Automation Competition (ARIAC) 2019.
- Designed a PID controller for Turtlebot 3.
- Modelled a UR5 arm with a Parallel Gripper in RViz.
- Teleoperated gesture-controlled robotic arm.
- Pick n Place transporter bot (**First Prize**, IIT Roorkee; **6th of 400 teams**, IIT Bombay Techfest).
- Smart Garden (**First Prize**, Texas Instruments Hardware Hackathon).

## CONFERENCE REVIEWS

- Workshop on Agricultural Robotics for a Sustainable Future, IROS [Reviewer].
- ICDL 2025 - IEEE International Conference on Development and Learning, September 16–19, 2025, Prague, Czechia [Reviewer].